Department of Computer Science and Engineering

PROJECT RESEARCH PROPOSAL

Deep Learning-Based Crop Disease Detection System for Cotton, Maize, Rice, Sugarcane, and Wheat

Submitted by

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Abstract

Crop diseases significantly affect agricultural productivity, crop quality, and farmers' income. Early and accurate identification of plant diseases can help farmers take appropriate preventive and corrective measures. This project proposes a deep learning-based Crop Disease Detection System that uses leaf images to identify diseases in five major crops: Cotton, Maize, Rice, Sugarcane, and Wheat.

The system uses a Convolutional Neural Network (CNN) to classify crop leaf images into disease and healthy categories. The collected image dataset will be organized into appropriate classes and preprocessed through image resizing, normalization, and data augmentation where necessary. The model will be trained and evaluated using separate training, validation, and testing datasets. Performance will be assessed using accuracy, precision, recall, F1-score, and a confusion matrix to evaluate classification effectiveness and identify potential weaknesses, including class imbalance.

The proposed system aims to provide an accessible interface where users can upload a crop leaf image and receive the predicted disease class along with the model's confidence score. The system may also present basic disease-related information to support agricultural decision-making. The project will document the model's performance, limitations, and suitability for practical use. The predictions are intended as preliminary decision-support information and should not replace expert agricultural diagnosis.

Keywords: Crop disease detection, deep learning, Convolutional Neural Network (CNN), image classification, precision agriculture, crop health monitoring.

Proposed System Flow Diagram

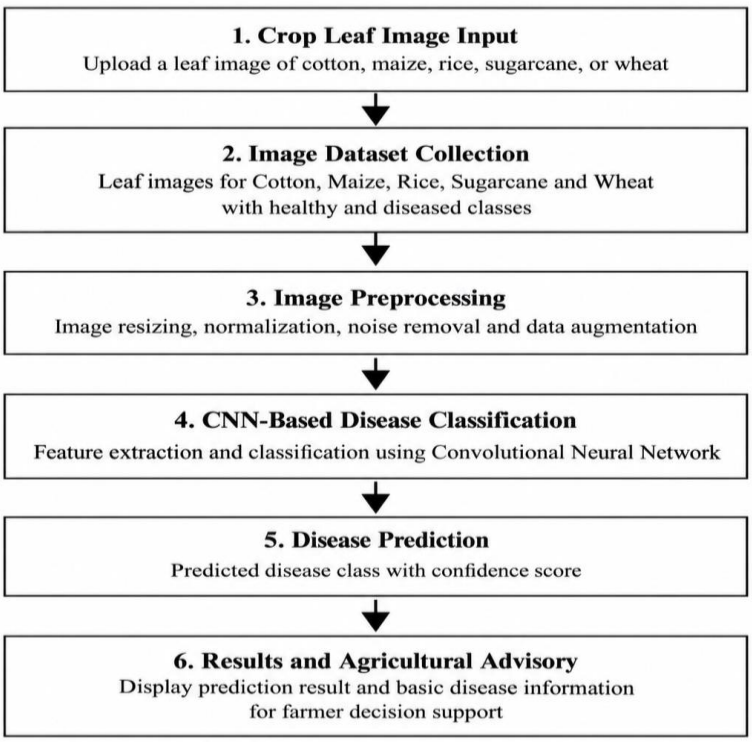


Figure 1. Proposed workflow for deep-learning-based crop disease detection and advisory.

Expected Outcomes

The proposed project is expected to develop a deep learning-based crop disease detection system capable of classifying leaf images of Cotton, Maize, Rice, Sugarcane, and Wheat into their respective healthy and diseased categories. The system will use a Convolutional Neural Network (CNN) to learn relevant visual features and predict the corresponding disease class from an input image.

The project is expected to provide a user-friendly interface for uploading leaf images and displaying the predicted class along with a confidence score. It will also evaluate the model using performance metrics such as accuracy, precision, recall, F1-score, and a confusion matrix. The evaluation will help identify classification errors and the effects of class imbalance on prediction performance.

The final outcome will be a working prototype that demonstrates the application of deep learning in crop disease identification, provides basic disease-related information to support farmers, and documents the model's performance and limitations. The system's predictions will serve as preliminary decision-support information and will not replace professional agricultural diagnosis.